SIMATS ENGINEERING

ASSESSMENT TOOL 2

Field	Details
Student Name	Dhakshitha A
Register Number	1925260126
Course Name	Computer Networks
Course Code	CSA0709
Course Outcome Covered	CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology (BL3)
Assessment Tool Weightage	30%
Total Marks	5 questions x 10 marks (Annexure A: Scenario-Based)

Code of Conduct

I, Dhakshitha A / 1925260126 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Criteria	Weightage	Q1	Q2	Q3	Q4	Q5
Understanding of Scenario	20%					
Identification of Key Issues	20%					
Application of Concepts	25%					
Decision Making	20%					
Clarity of Response	15%					

Question 1 (10 Marks) - Smart Campus Subnetting (192.168.100.0/24)

Question: A newly established Smart Campus is deploying a secure network. The administrator has the block 192.168.100.0/24 with masks already assigned: School of Computing /25, School of Electronics /26, School of Mechanical Engineering /27. Determine (1) network and broadcast address of each department, (2) valid host range, (3) number of usable hosts, (4) remaining unused range.

Understanding of the Scenario

The /24 block has 256 addresses (192.168.100.0 - 192.168.100.255). Three departments need different sizes, so Variable Length Subnet Masking (VLSM) is used. The largest subnet is placed first to avoid overlap and keep each subnet on a proper block boundary.

Key Concepts and Formulas

- Block size = $2^{(32 - \text{prefix})}$: /25 = 128, /26 = 64, /27 = 32
- Usable hosts = $2^{h} - 2$ (h = host bits); network and broadcast addresses are not assignable
- /25 = 255.255.255.128, /26 = 255.255.255.192, /27 = 255.255.255.224

Solution

Department	Subnet / Mask	Network Address	Broadcast Address	Valid Host Range	Usable Hosts
School of Computing	192.168.100.0/25 255.255.255.128	192.168.100.0	192.168.100.127	192.168.100.1 - 192.168.100.126	$2^7 - 2 = 126$
School of Electronics	192.168.100.128/26 255.255.255.192	192.168.100.128	192.168.100.191	192.168.100.129 - 192.168.100.190	$2^6 - 2 = 62$
School of Mechanical Engg.	192.168.100.192/27 255.255.255.224	192.168.100.192	192.168.100.223	192.168.100.193 - 192.168.100.222	$2^5 - 2 = 30$

Remaining (Unused) Address Range

Addresses used = 128 + 64 + 32 = 224. Remaining = 256 - 224 = 32 addresses.
Unused range: 192.168.100.224 - 192.168.100.255 (i.e., 192.168.100.224/27; network 192.168.100.224, broadcast 192.168.100.255, 30 usable hosts), which is kept reserved for future expansion.

Key Issues and Decision

- Subnets must not overlap: Computing .0-.127, Electronics .128-.191, Mechanical .192-.223.
- Allocating from the largest to the smallest subnet keeps every network address aligned to its block size.
- Each department has enough hosts for its needs while the unused /27 gives room for growth, so the design is efficient and scalable.

Question 2 (10 Marks) - Multinational Company Subnetting (10.10.0.0/16)

Question: A multinational software company is assigned the private block 10.10.0.0/16. Masks allocated: Software Development /24, Quality Assurance /25, Human Resources /26, Executive Management /27. Determine (1) network and broadcast address, (2) valid host range, (3) usable hosts, (4) unused range remaining.

Understanding of the Scenario

A /16 block holds 2^{16} = 65,536 addresses (10.10.0.0 - 10.10.255.255). Four departments get subnets of different sizes (VLSM), allocated in descending order of size, one after another with no overlap.

Key Concepts and Formulas

- Block sizes: /24 = 256, /25 = 128, /26 = 64, /27 = 32 addresses
- Usable hosts = 2^h - 2
- Masks: /24 = 255.255.255.0, /25 = .128, /26 = .192, /27 = .224

Solution

Department	Subnet / Mask	Network Address	Broadcast Address	Valid Host Range	Usable Hosts
Software Development	10.10.0.0/24 255.255.255.0	10.10.0.0	10.10.0.255	10.10.0.1 - 10.10.0.254	2^8 - 2 = 254
Quality Assurance	10.10.1.0/25 255.255.255.128	10.10.1.0	10.10.1.127	10.10.1.1 - 10.10.1.126	2^7 - 2 = 126
Human Resources	10.10.1.128/26 255.255.255.192	10.10.1.128	10.10.1.191	10.10.1.129 - 10.10.1.190	2^6 - 2 = 62
Executive Management	10.10.1.192/27 255.255.255.224	10.10.1.192	10.10.1.223	10.10.1.193 - 10.10.1.222	2^5 - 2 = 30

Remaining (Unused) Address Range

Addresses used = 256 + 128 + 64 + 32 = 480. Remaining = 65,536 - 480 = 65,056 addresses.
Unused range: 10.10.1.224 - 10.10.255.255.

Key Issues and Decision

- Software Development is placed first (largest), so QA, HR and Executive follow contiguously inside 10.10.1.0/24 without overlap.
- The very large unused space (10.10.1.224 onward) can be used for new departments, servers, WAN links or further VLSM subnets.
- Private addressing (10.0.0.0/8 range) is suitable for internal use; NAT is needed for Internet access.

Question 3 (10 Marks) - Smart Hospital IPv4 to IPv6 Migration

Question: A smart hospital is migrating from IPv4 to IPv6 for thousands of medical devices and IoT sensors. Assigned network: 2001:0db8:abcd:0000:0000:0000:0000:0000/64. Perform: (1) compress the address, (2) expand the compressed address, (3) identify the network prefix, (4) calculate the number of host addresses in the subnet, (5) total number of possible host addresses in this network.

Understanding of the Scenario

IPv4 (32-bit, about 4.3 billion addresses) cannot support the huge number of connected medical devices and sensors. IPv6 uses 128-bit addresses written as eight groups of 16-bit hexadecimal values. A /64 prefix means the first 64 bits identify the network and the last 64 bits identify the interface (host).

Step 1: Compressed Notation

Rules: (a) drop leading zeros in each group; (b) replace one longest run of all-zero groups with '::' (only once).
Full: 2001:0db8:abcd:0000:0000:0000:0000:0000/64
After dropping leading zeros: 2001:db8:abcd:0:0:0:0:0/64
Compressed: 2001:db8:abcd::/64

Step 2: Expansion of the Compressed Address

2001:db8:abcd:: has 3 explicit groups, so '::' stands for 8 - 3 = 5 zero groups. Each group is padded to 4 hex digits.
Expanded: 2001:0db8:abcd:0000:0000:0000:0000:0000/64

Step 3: Network Prefix

/64 means the first 64 bits (first four groups) are the network prefix.
Network prefix: 2001:0db8:abcd:0000 (2001:db8:abcd:0::/64)
Interface ID (host portion): remaining 64 bits, range ::0 to ::ffff:ffff:ffff:ffff.

Steps 4 and 5: Number of Host Addresses

Item	Value
Total bits / prefix bits	128 / 64
Host (interface ID) bits	128 - 64 = 64
Addresses in the subnet	2^{64} = 18,446,744,073,709,551,616 (about 1.8×10^{19})
Note on usable hosts	IPv6 has no broadcast address, so there is no '-2' deduction as in IPv4. By convention the all-zero interface ID (subnet-router anycast) is reserved, so about 2^{64} - 1 addresses are assignable to devices.

Key Issues and Decision

- Address exhaustion in IPv4 and the need for a huge number of unique device addresses: IPv6 solves this.
- A /64 is the standard subnet size for a LAN (required for SLAAC autoconfiguration), so it is the correct choice for the hospital's device networks.
- A dual-stack approach (IPv4 and IPv6 together) is recommended during migration so existing medical systems keep working.

Question 4 (10 Marks) - ISP Core Router Packet Forwarding

Question: An ISP core router has routing table entries 192.168.1.0/24, 192.168.2.0/24 and 192.168.3.0/24. A packet arrives with destination IP 192.168.2.45. Determine (1) the destination subnet, (2) the matching routing table entry, (3) the next-hop network/interface used to forward the packet, (4) the default route to use if there is no match.

Understanding of the Scenario

A router forwards each packet by comparing the destination IP with its routing table using the **longest prefix match** rule. The destination address is ANDed with each entry's subnet mask; the entry that matches with the longest prefix is selected.

Step 1 and 2: Destination Subnet and Matching Entry

Routing Entry	Address Range	Does 192.168.2.45 match?
192.168.1.0/24	192.168.1.0 - 192.168.1.255	No (third octet 2 is not 1)
192.168.2.0/24	192.168.2.0 - 192.168.2.255	**Yes** (third octet = 2)
192.168.3.0/24	192.168.3.0 - 192.168.3.255	No (third octet 2 is not 3)

Binary check: 192.168.2.45 AND 255.255.255.0 = 192.168.2.0, which equals the network address of 192.168.2.0/24.
Destination subnet: 192.168.2.0/24 (network 192.168.2.0, broadcast 192.168.2.255, hosts 192.168.2.1 - 192.168.2.254; 192.168.2.45 is a valid host).
Matching routing entry: 192.168.2.0/24.

Step 3: Next-Hop Network / Interface

The table lists only the destination networks, so the next hop is the one associated with the matching entry: the router forwards the packet out of the interface (or to the next-hop router) that leads to the 192.168.2.0/24 customer network, for example Interface Gi0/1 (assumed name) to customer network 192.168.2.0/24. The packet is not sent toward the 192.168.1.0/24 or 192.168.3.0/24 interfaces. Because the network is reached through that interface, the router ARPs for 192.168.2.45 and delivers the frame there.

Step 4: Default Route

If the destination matches none of the entries, the router uses the default route **0.0.0.0/0** (mask 0.0.0.0), which has the shortest possible prefix and matches every address. It points to the ISP's upstream/Internet gateway, e.g. `ip route 0.0.0.0 0.0.0.0 <upstream-next-hop>`. If no default route exists, the packet is dropped and an ICMP 'destination unreachable' message is returned.

Key Issues and Decision

- Longest prefix match decides the route when entries overlap; here only one /24 matches, so the decision is clear.
- A default route is essential on an ISP router for traffic to unknown or external destinations.
- Packet flow: arrives -> lookup -> match 192.168.2.0/24 -> forward via its interface -> delivered to 192.168.2.45.

Question 5 (10 Marks) - Two LANs Connected Through a Layer-3 Router

Question: Two departments are connected through a Layer-3 router. Administration LAN 192.168.10.0/26 (gateway 192.168.10.1) and Finance LAN 192.168.10.64/26 (gateway 192.168.10.65). Determine (1) network and broadcast address for each LAN, (2) valid host range, (3) suitable IP addresses for three computers in each LAN, (4) the default gateway for hosts in each LAN.

Understanding of the Scenario

The block 192.168.10.0 is divided into two /26 subnets. Each /26 has $2^6 = 64$ addresses (mask 255.255.255.192) and 64 - 2 = 62 usable hosts. Hosts in different LANs can communicate only through the Layer-3 router, whose interface in each LAN acts as the default gateway.

Steps 1 and 2: Network, Broadcast and Host Range

LAN	Network Address	Broadcast Address	Valid Host Range	Usable Hosts
Administration 192.168.10.0/26	192.168.10.0	192.168.10.63	192.168.10.1 - 192.168.10.62	62
Finance 192.168.10.64/26	192.168.10.64	192.168.10.127	192.168.10.65 - 192.168.10.126	62

Step 3: Suggested Host IP Addresses

The first address in each range is used by the router interface (gateway), so hosts are given addresses after it. Subnet mask for all hosts: 255.255.255.192.

LAN	Device	IP Address	Subnet Mask	Default Gateway
Administration	PC1	192.168.10.2	255.255.255.192	192.168.10.1
Administration	PC2	192.168.10.3	255.255.255.192	192.168.10.1
Administration	PC3	192.168.10.4	255.255.255.192	192.168.10.1
Finance	PC4	192.168.10.66	255.255.255.192	192.168.10.65
Finance	PC5	192.168.10.67	255.255.255.192	192.168.10.65
Finance	PC6	192.168.10.68	255.255.255.192	192.168.10.65

Step 4: Default Gateways

- **Administration LAN hosts:** default gateway = 192.168.10.1 (router interface in Administration LAN).
- **Finance LAN hosts:** default gateway = 192.168.10.65 (router interface in Finance LAN).

Key Issues and Decision

- Every host must be in the correct subnet and use its own LAN's gateway; a wrong gateway would block inter-LAN traffic.
- Gateway addresses (.1 and .65) lie within the valid host ranges and are not the network or broadcast addresses, and no two devices share an IP.
- Example: PC1 (192.168.10.2) pinging PC4 (192.168.10.66) sends the packet to 192.168.10.1; the router forwards it out of the Finance interface (192.168.10.65) to PC4.
- Simulation (Cisco Packet Tracer): configure the router interfaces with the gateway IPs and mask 255.255.255.192, set the PC IP/mask/gateway, and verify with ping.

- End of Assessment -